

SEMAPHORES

Niranjan Kumar

241801177

Write a C program to implement the **Producer–Consumer problem** using semaphores. The program should simulate a bounded buffer where a producer produces items and a consumer consumes items.

The program must use the following variables:

- `mutex` → to ensure mutual exclusion
- `full` → to count filled slots
- `empty` → to count empty slots
- `x` → to track number of items

The user should be able to choose:

1. Producer
2. Consumer
3. Exit

Sample Input 1:

1
1
2
3

Sample Output 1:

----- Producer Consumer Problem -----

1. Producer
2. Consumer
3. Exit

Enter your choice: 1 Producer produces item 1

Enter your choice: 1 Producer produces item 2

Enter your choice: 2 Consumer consumes item 2

Enter your choice: 3 Exiting program... Final buffer items produced: 1

Sample Input 2:

2

1

3

Sample Output 2:

----- Producer Consumer Problem -----

Enter your choice: 2 Buffer is empty!! Consumer cannot consume.

Enter your choice: 1 Producer produces item 1

Enter your choice: 3 Exiting program... Final buffer items produced: 1

CODE:

// Producer Consumer

#include <stdio.h>

#include <stdlib.h>

int mutex = 1, full = 0, empty = 5, x = 0;

int wait(int s)

{

return --s;

```
}
```

```
int signal(int s) {
```

```
    return ++s;
```

```
}
```

```
void producer()
```

```
{
```

```
    mutex = wait(mutex);
```

```
    empty = wait(empty);
```

```
    full = signal(full);
```

```
    x++;
```

```
    printf("\nProducer produces item %d", x);
```

```
    mutex = signal(mutex);
```

```
}
```

```
void consumer()
```

```
{
```

```
    mutex = wait(mutex);
```

```
    full = wait(full);
```

```
    empty = signal(empty);
```

```
printf("\nConsumer consumes item %d", x);
```

```
x--;
```

```
mutex = signal(mutex);
```

```
}
```

```
int main()
```

```
{ int
```

```
n;
```

```
printf("\n----- Producer Consumer Problem -----");
```

```
printf("\n1. Producer"); printf("\n2. Consumer");
```

```
printf("\n3. Exit");
```

```
while(1)
```

```
{
```

```
printf("\nEnter your choice: ");
```

```
scanf("%d",&n);
```

```
switch(n)
```

```
{
```

```
case 1:
```

```
    if(mutex==1 && empty!=0)
    {
        producer();
    }
else
    {
        printf("\nBuffer is full!! Producer cannot produce.");
    }
    break;

case 2:
    if(mutex==1 && full!=0)
    {
        consumer();
    }
else
    {
        printf("\nBuffer is empty!! Consumer cannot consume.");
    }
    break;
```

case 3:

printf("\nExiting program...");

printf("\nFinal buffer items produced: %d\n", x);

exit(0);

default:

printf("\nInvalid choice. Try again.");

}

}

return 0;

}